STT 861: Theory of Probability & Statistics I

Lecture 1: Course Introduction & Set Theory I

Guanqun Cao

Department of Statistics & Probability
Michigan State University

Sep 1, 2026

Reference: Casella & Berger Ch. 1.1

Lecture 1 Outline

Welcome to STT 861

What this course covers:

- Chapter 1: Probability Theory
- Chapter 2: Transformations & Expectations
- Chapter 3: Common Families of Distributions
- Chapter 4: Multiple Random Variables
- Chapter 5: Properties of a Random Sample

After this course you will:

- Rigorously prove probability theorems
- Work fluently with distributions and expectations
- Understand convergence concepts (LLN, CLT)
- Read and write statistical theory proofs

Welcome to STT 861

Course Information

Instructor: Guanqun Cao
Email: caoguanq@msu.edu

Time: T/Th 11:00–12:20 pm
Room: Wells Hall C506

Text: Casella & Berger,
Statistical Inference, 2nd ed.

Platform: D2L (grades)
Course website: (slides, HW)
Communication: MSU Email

Something about me

- Associate professor in the Department of Statistics & Prob
- Got my PhD from MSU in 2012
- On the faculty at Auburn University from 2012-2023
- Interested in statistical methodology (imaging classification, high dimension data, functional data) and cool applications (biomedical engineering, social economics, etc)

Two Truths and a Lie

- I ride motorcycle to work.
- Since graduate school, I've never lived in a house for more than 5 years.
- I used to be a national champion in martial arts.

Something about You

- Your name, program and year
- Your academic and non-academic interests
- What is your goal in this course
- Anything else you want me to know (e.g., your two truths and a lie)

Course Overview

Probability and Distribution Theory

- Foundation of mathematical statistics
- Required for 1st-year MS students in Statistics
- Prerequisites: calculus
- The goal is to introduce basic definitions and theories in probability and statistics
- The course will involve derivation, proof, and calculation

Grading Policy

Grade Components:

Component	Weight
Homework	20%
Exams I & II (combined)	40%
Final Exam	20%
In-class pop quizzes	20%

Grade Scale (4.0)

4.0: $\geq 90\%$	2.0: 70–75%
3.5: 85–90%	1.5: 65–70%
3.0: 80–85%	1.0: 60–65%
2.5: 75–80%	0.0: $< 60\%$

Key Policies:

- Attendance is mandatory
- Penalty for late HW
- One lowest HW dropped
- One lowest in-class quiz dropped

Important Dates

Date	**Event**
August 31, 2026	Classes begin
October 15	**Exam I** (Chapters 1–2)
October 27	No class (Fall break)
November 19	**Exam II** (Chapters 3–4)
November 26	No class (Thanksgiving)
December 10	**Final Exam**

Course Expectations

Be an active participant — statistics is not a spectator sport!

- Come to class prepared; read the relevant CB section first
- Ask questions — when something is unclear, just ask
- Engage with in-class activities; help your peers learn

Why Study Probability?

The Fundamental Goal

Objective of a statistician: To draw conclusions about a *population* of objects by conducting an *experiment*.

$$\text{Set Theory} \longrightarrow \text{Probability Theory} \longrightarrow \text{Statistics}$$

Each layer builds on the foundation of the previous.

Probability is the language of uncertainty.

For example:

- What is the chance I will win Mega Millions?
- What are the odds of meeting your life partner in Michigan?

In many applications, probability simply describes the relative frequency at which events will occur.

Objective of a statistician: To draw conclusions about a population of objects by conducting an experiment.

- Statistics builds upon the foundation of Probability Theory.
- Probability Theory builds upon the foundation of Set Theory.
    - sample space
    - outcomes
    - events

Why Study Probability?

Gambling & Decision Problems:

- Three friends toss a coin – odd man out pays dinner. What is the chance *you* pay?
- **Monty Hall Problem:** Three doors, one car. Should you switch?
- **100 Prisoners Dilemma:** Can they all find their numbers using a cyclic strategy?

Real-World Applications:

- Medical screening: If a test is positive, what is the probability you have the disease?
- Epidemiology: Is smoking independent of lung cancer?
- Machine learning: Bayesian inference, probabilistic models
- Finance: Risk quantification, option pricing

The Birthday Problem – A Motivating Puzzle

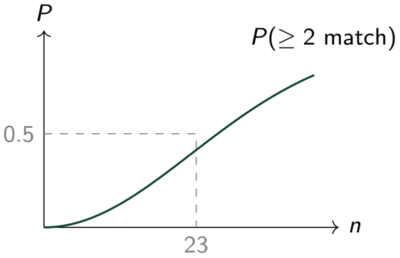
Question

In a room of n people, what is the probability that *at least two* share a birthday?

Most people guess this probability is very small for $n = 23$.

The answer: $\approx 50.7\%$**!**

- With $n = 57$: probability exceeds 99%
- We use the *complement*:
  $P(\text{at least 2 match}) = 1 - P(\text{all distinct})$
- $P(\text{all distinct}) = \frac{365}{365} \cdot \frac{364}{365} \cdots \frac{365-n+1}{365}$



Solving this precisely requires the counting tools from Section 1.2.

The Monty Hall Problem

Game Show Scenario

Three doors: behind one is a car, behind the others are goats.

1. You pick a door (say Door 1).
2. The host (who knows what's behind each door) reveals a goat behind one of the other two doors.
3. You are offered the chance to switch. **Should you?**

Intuition (wrong): "50-50, doesn't matter."

Correct answer: Always switch!

By switching: $P(\text{win}) = 2/3$.
By staying: $P(\text{win}) = 1/3$.

Car	Stay	Switch
Door 1	Win	Lose
Door 2	Lose	Win
Door 3	Lose	Win

Each row equally likely (prob. $1/3$).
Switching wins in 2 out of 3 cases.

Sample Space

Definition

The **sample space** S is the set of *all* possible outcomes of a random experiment.

Some Examples:

- Outcome of a sick patient receiving medication:

$$S = \{\text{Success}, \text{ Failure}\}$$

- An urn contains 4 white balls and 2 red balls. Two balls are drawn (without replacement). What is the sample space?

Sample Space

Definition

The **sample space** S is the set of *all* possible outcomes of a random experiment.

Some Examples:

- Outcome of a sick patient receiving medication:

$$S = \{\text{Success}, \text{ Failure}\}$$

- An urn contains 4 white balls and 2 red balls. Two balls are drawn (without replacement). What is the sample space? If only 1 red ball is in the urn, how does the sample space change?

Sample Space – More Examples

Coin Tossing:

No. of Tosses	Sample Space
1	$\{H, T\}$
2	$\{HH, HT, TH, TT\}$
3	$\{HHH, THH, HTH, HHT,$ $TTT, HTT, THT, TTH\}$
n	2^n equally-likely outcomes

Sampling for smokers:

- Sample 100 people to estimate the proportion who smoke.
- The sample space is:

$$S = \{0, 1, 2, \ldots, 100\}$$

where each element represents the number of smokers in the sample.

Discrete and Non-Discrete Sample Spaces

Discrete Sample Space

A sample space S is called **discrete** if it contains a *finite* number of sample points, or if it contains an *infinite* number of points that can be placed in 1-to-1 correspondence with the positive integers.

- A **finite** sample space has finitely many points.
- A **countably infinite** sample space has infinitely many points indexed by $\mathbb{N}$.

Example – Toss a coin until the first Head occurs:

$$S = \{H,\ TH,\ TTH,\ TTTH,\ \ldots\}$$
$$S = \{x : x = 1,\ 2,\ 3,\ \ldots\} \quad \text{(countably infinite)}$$

Continuous (Uncountably Infinite) Sample Spaces

A sample space is called **uncountably infinite** if it contains an infinite number of points that *cannot* be placed in 1-to-1 correspondence with the integers.

Continuous sample spaces are the most important examples:

$$\mathbb{R} = (-\infty, \infty), \qquad \mathbb{R}^+ = (0, \infty), \qquad [0, \infty)$$

Examples where $S = \mathbb{R}^+$:

- Age of an individual
- Survival time of a patient
- A person's height or weight

Remark

For continuous S, we cannot enumerate the outcomes. Probabilities are assigned to *intervals* (or more generally, Borel sets) rather than to individual points.

Outcome and Event

Definitions

- **Event:** any collection of possible outcomes of an experiment, i.e. a *subset* of the sample space S (including S itself).
- **Sample point:** an event that has no proper subsets (a single outcome).
- **Null event** $\emptyset$: an event containing *no* sample points.

Remarks:

1. An event can be made up of one *or more* sample points.
2. Strictly speaking, events are the subsets of S for which a probability is defined (i.e. members of a σ-algebra $\mathcal{B}$ on S).

Examples of Events

Coin Tossing

Toss a fair coin twice: $S = \{HH, HT, TH, TT\}$.

Event $E =$ "at least one Head in two tosses":

$$E = \{HT,\ TH,\ HH\}$$

Urn Sampling

An urn contains Red (R) and White (W) balls. Two balls are chosen *without* replacement.

Event $E =$ "exactly one White ball is drawn":

$$E = \{WR,\ RW\}$$

Note that order matters here because the draws are sequential.

When Does an Event Occur?

Let A be an event, i.e. a subset of S.

We say the event A **occurs** if the outcome of the experiment is in the set A.

When speaking of probability we generally refer to the *probability of an event*, rather than of a set – but we use the two terms interchangeably.

Illustration

Roll a fair die: $S = \{1, 2, 3, 4, 5, 6\}$.

Let $A = \{2, 4, 6\}$ ("even number").

If the die shows 4, then $4 \in A$, so the event A *occurs*.
If the die shows 3, then $3 \notin A$, so A does *not* occur.

Relationships Between Sets

Containment and Equality

1. **Containment (subset):**
$$A \subset B \iff x \in A \Rightarrow x \in B$$
"If A occurs, then B must also occur."
2. **Equality:**
$$A = B \iff A \subset B \text{ and } B \subset A$$

Die Roll

$S = \{1, 2, 3, 4, 5, 6\}$, $A = \{6\}$ ("six"), $B = \{4, 5, 6\}$ ("more than three").

$A \subset B$: rolling a six implies rolling more than three.
$B \not\subset A$: rolling a five does not imply rolling a six.

Set Operations

Let A and B be two events on S.

Three Basic Operations

Union ($A \cup B$ or $A + B$):

$$A \cup B = \{x : x \in A \ \text{ or } \ x \in B\}$$

Intersection ($A \cap B$ or AB):

$$A \cap B = \{x : x \in A \ \text{ and } \ x \in B\}$$

Complementation (A^c or $\bar{A}$ or A'):

$$A^c = \{x \in S : x \notin A\}$$

Note: $S^c = \emptyset \quad \text{and} \quad \emptyset^c = S$.

Example: Numbered Balls in an Urn

Set Operations in Practice

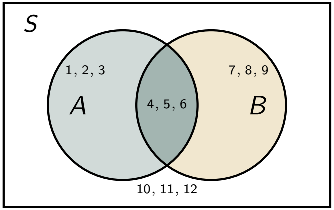
An urn contains 12 balls numbered $1, 2, \ldots, 12$. One ball is drawn at random.

$$S = \{1,\ 2,\ \ldots,\ 12\}$$

Suppose

$$A = \{1, 2, 3, 4, 5, 6\}, \qquad B = \{4, 5, 6, 7, 8, 9\}.$$

$$A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$
$$A \cap B = \{4, 5, 6\}$$
$$A^c = \{7, 8, 9, 10, 11, 12\}$$



Algebraic Laws of Set Operations

The Laws

Commutative:
$A \cup B = B \cup A$
$A \cap B = B \cap A$

Associative:
$A \cup (B \cup C) = (A \cup B) \cup C$
$A(BC) = (AB)C$

Distributive:
$A(B \cup C) = AB \cup AC$
$A \cup (BC) = (A \cup B)(A \cup C)$

Idempotency:
$A \cup A = A, \quad AA = A$

De Morgan's Laws:
$(A \cup B)^c = A^c \cap B^c$
$(AB)^c = A^c \cup B^c$

Proof strategy for De Morgan:

To show $X = Y$ we show $X \subset Y$ *and* $Y \subset X$.

Proof of $(A \cup B)^c = A^c \cap B^c$:

($\subset$**)** Let $x \in (A \cup B)^c$.
$\Rightarrow x \notin A \cup B$
$\Rightarrow x \notin A$ *and* $x \notin B$
$\Rightarrow x \in A^c \cap B^c$. ✓

($\supset$**)** Let $x \in A^c \cap B^c$.
$\Rightarrow x \notin A$ *and* $x \notin B$
$\Rightarrow x \notin A \cup B$
$\Rightarrow x \in (A \cup B)^c$. ✓ $\square$

Set Operations – Worked Example

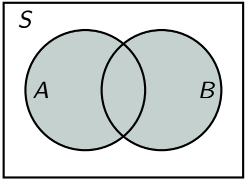
Numbered Balls in an Urn

12 balls numbered $1, 2, \ldots, 12$. Draw one ball.
$S = \{1, 2, 3, \ldots, 12\}$

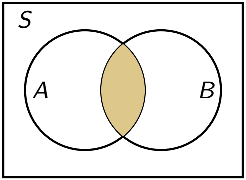
$A = \{1, 2, 3, 4, 5, 6\}$ (“draw a number ≤ 6”)
$B = \{4, 5, 6, 7, 8, 9\}$ (“draw a number between 4 and 9”)

Operation	**Result**
$A \cup B$	$\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$
$A \cap B$	$\{4, 5, 6\}$
A^c	$\{7, 8, 9, 10, 11, 12\}$
$B - A$	$\{7, 8, 9\}$
$A - B$	$\{1, 2, 3\}$

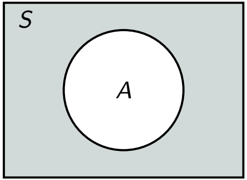
Venn Diagrams



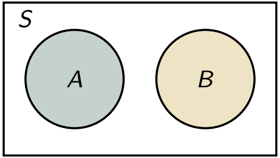
(a) $A \cup B$ (shaded)



(b) $A \cap B$ (shaded)



(c) A^c (shaded)



(d) Mutually exclusive: $A \cap B = \emptyset$

Symmetric Difference and Three-Set Venn Diagrams

Symmetric Difference

$B \triangle A = (B - A) \cup (A - B)$
$= (A \cup B) - (A \cap B)$

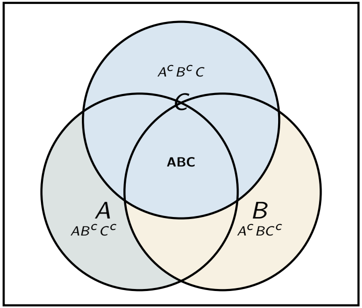
Elements in exactly one of A, B.

Example

$A = \{1, 2, 3\},\ B = \{2, 3, 4, 5\}$

$A \triangle B = \{1, 4, 5\}$

Three sets A, B, C**:** $2^3 = 8$ disjoint regions.



Infinite Collections of Sets

Infinite Unions and Intersections

Let $A_1, A_2, A_3, \ldots$ be events defined on the same sample space S. Then:

$$\bigcup_{i=1}^{\infty} A_i = \{x \in S : x \in A_i \text{ for some } i\}$$

$$\bigcap_{i=1}^{\infty} A_i = \{x \in S : x \in A_i \text{ for all } i\}$$

More generally, for any index set Γ and events A_α, $\alpha \in \Gamma$: $\bigcup_{\alpha \in \Gamma} A_\alpha = \{x \in S : x \in A_\alpha \text{ for some } \alpha\}$

Example — Infinite Unions and Intersections

Example

Let $S = (0, 1]$ and define $A_i = \left[\frac{1}{i}, 1\right]$. Then:

$$\bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^{\infty} \left[\frac{1}{i}, 1\right] = \left\{x \in (0, 1] : x \in \left[\frac{1}{i}, 1\right] \text{ for some } i\right\} = \{x \in (0, 1]\} = (0, 1]$$

$$\bigcap_{i=1}^{\infty} A_i = \bigcap_{i=1}^{\infty} \left[\frac{1}{i}, 1\right] = \left\{x \in (0, 1] : x \in \left[\frac{1}{i}, 1\right] \text{ for all } i\right\} = \{x \in (0, 1] : x \in [1, 1]\} = \{1\}$$

Disjoint Events and Partitions

Disjoint (Mutually Exclusive)

A and B are **disjoint** (mutually exclusive) if $A \cap B = \emptyset$.

More generally, $A_1, A_2, \ldots$ are **pairwise disjoint** if $A_i \cap A_j = \emptyset$ for all $i \neq j$.

Partition

Events $A_1, A_2, \ldots$ form a **partition of S** if:

1. They are pairwise disjoint: $A_i \cap A_j = \emptyset$ for $i \neq j$
2. They cover S: $\bigcup_{i=1}^{\infty} A_i = S$

Disjoint Events and Partitions

Die Roll Partition

$A_1 = \{1, 3, 5\}$ (odd), $A_2 = \{2, 4, 6\}$ (even).
$A_1 \cap A_2 = \emptyset$; $A_1 \cup A_2 = S$. ✓

Also: $\{A, A^c\}$ always partitions S.

Interval Partition

$A_i = [i, i + 1)$, $i = 0, 1, 2, \ldots$

Pairwise disjoint: $A_i \cap A_j = \emptyset$ for $i \neq j$.✓
$\bigcup_{i=0}^{\infty} A_i = [0, \infty)$.✓

Solving the Odd-Man-Out Dinner Problem

Three Friends Tossing Coins

You (Y), Friend 1 (F_1), Friend 2 (F_2) each toss a fair coin.
The "odd man out" pays (the one whose result differs from the other two).

Sample Space:

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}, \quad |S| = 8$$

All 8 equally likely with probability 1/8 each.

Event "You (Y) pay" = {Y gets H, both friends get T} ∪ {Y gets T, both friends get H}:

$$A = \{HTT,\ THH\}, \quad P(A) = \frac{2}{8} = \frac{1}{4}$$

By symmetry, each friend also pays with probability 1/4.

Solving the Odd-Man-Out Dinner Problem

Note

With probability $1 - 3 \times \frac{1}{4} = \frac{1}{4}$, *no one* is the odd man out (everyone got the same result), so no one pays. In practice, they'd re-toss!

In-Class Activity – Lecture 1

Activity 1 – Set Theory

A six-sided die is rolled. $S = \{1, 2, 3, 4, 5, 6\}$. Define:

$$A = \{2, 4, 6\}, \quad B = \{1, 2, 3\}, \quad C = \{3, 4, 5, 6\}$$

- (a) Find $A \cup B$, $A \cap C$, and B^c.
- (b) Find $(A \cup B)^c$ and verify DeMorgan's law: $(A \cup B)^c = A^c \cap B^c$.
- (c) Find $B \triangle C$ (symmetric difference).
- (d) Do $\{B,\ A \cap C,\ B^c \cap C^c\}$ form a partition of S?
  Justify: show pairwise disjoint and union equals S.
- (e) **Bonus:** Three events A, B, C are defined on $S = \{1, \ldots, 8\}$. $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, $C = \{5, 6, 7, 8\}$.
  Find $A \cap (B \cup C)$ and verify via distributive law.

Activity 1 – Solutions

$S = \{1,2,3,4,5,6\},\ \ A = \{2,4,6\},\ \ B = \{1,2,3\},\ \ C = \{3,4,5,6\}$

(a) $A \cup B = \{1,2,3,4,6\}; \quad A \cap C = \{4,6\}; \quad B^c = \{4,5,6\}$

(b) $(A \cup B)^c = \{5\}.\ \ A^c = \{1,3,5\},\ B^c = \{4,5,6\},\ A^c \cap B^c = \{5\}.\checkmark$ DeMorgan verified.

(c) $B \triangle C = (B - C) \cup (C - B) = \{1,2\} \cup \{4,5,6\} = \{1,2,4,5,6\}$

(d) Check pairwise disjoint:
$B \cap (A \cap C) = \{1,2,3\} \cap \{4,6\} = \emptyset\checkmark; \quad B \cap (B^c \cap C^c) = \emptyset\checkmark;$
$(A \cap C) \cap (B^c \cap C^c) = \{4,6\} \cap \{4,5\}^c = \emptyset\ldots$
Check union: $\{1,2,3\} \cup \{4,6\} \cup \{5\} = \{1,2,3,4,5,6\} = S\checkmark.$
Yes, it is a partition.

(e) $B \cup C = \{1,2,3,4,5,6,7,8\};\ A \cap (B \cup C) = \{2,4,6\}.$
Distributive: $(A \cap B) \cup (A \cap C) = \{2\} \cup \{4,6\} = \{2,4,6\}.\checkmark$